

How to perform a carbon footprint

Collect the
product input data
You!

Column E

e.g. 1040 kg_{steel}/V
6.2 kg/100 V-km

x 1/150,000km
x 1

Transform them per
functional unit
FU= 1 V-km You

Column F

0.0069 kg_{steel}/V-km
0.062 kg_{gas}/V-km

Find GHG emission
factors per unit input:
in database

Column A→ H

2.2 kg_{CO2eq}/kg_{steel}
4.45 kg_{CO2eq}/kg_{gas}

Multiply F x H
to get the carbon
footprint per FU

Column I

= 0.015 kg_{CO2eq}/FU
= 0.275 kg_{CO2eq}/FU

Process #	Transportation by Gas Vehicle Functional unit (FU)= 1 vehicle km Life Cycle stage (150,000 km lifespan)	Quantity original unit	Unit
Manufacturing (Materials + Processing)			
per vehicle over lifespan			
17299	steel, low-alloyed, hot rolled	1040	kg
16619	sheet rolling, steel	780	kg
458	aluminium, primary, ingot	55	kg
16616	Copper & other metals (Zn, Ni, Pb)	38	kg
12456	platinum	1.70E-03	kg
14632	polyethylene, polypropylene & PVC	175	kg
17718	synthetic rubber	46	kg
7943	glas	32	kg
7364	electronics, for control units	6.1	kg
7587	ethylene, paints, fluids	30	kg
17773	tap water	3400	kg
9148	heat, district or industrial, natural gas	2400	MJ
7235	electricity, medium voltage	2250	kWh
Transportation			
per vehicle			
18124	transport, freight train	556.50	metric ton*km
18241	transport, freight, lorry 16-32 metric ton	59.10	metric ton*km
18438	transport, freight, sea, container ship	11376.81	metric ton*km
Use phase			
per V-km			
14137	Supply chain petrol, low-sulfur	0.0619	kg
100001	Direct combustion	0.0619	kg petrol
Disposal			
per vehicle over lifespan			
19923	plastic mixture incineration	221.00	kg
19515	Glass to landfill	283.00	kg
7235	Avoided electricity plastic incineration	221.00	kWh

Quantity per FU
perFU
6.93E-03
5.20E-03
3.67E-04
2.53E-04
1.13E-08
1.17E-03
3.07E-04
2.13E-04
4.07E-05
2.00E-04
2.27E-02
1.60E-02
1.50E-02
perFU
3.71E-03
3.94E-04
7.58E-02
perFU
6.19E-02
6.19E-02
perFU
1.47E-03
1.89E-03
-0.00435

Process #	Carbon footprint [kgCO2equ/unit]	Carbon footprint [CO2equ./FU]
17299	2.18	1.51E-02
16619	0.31	1.62E-03
458	7.21	2.64E-03
16616	0.48	1.22E-04
12456	69669.94	7.90E-04
14632	2.33	2.72E-03
17718	2.97	9.10E-04
7943	1.06	2.25E-04
7364	31.86	1.30E-03
7587	1.45	2.89E-04
17773	0.00	9.87E-06
9148	0.08	1.21E-03
7235	0.38	5.77E-03
18124	0.05	1.70E-04
18241	0.19	7.52E-05
18438	0.01	7.71E-04
14137	1.02	6.30E-02
100001	3.43	2.12E-01
19923	2.35	3.46E-03
19515	0.01	1.07E-05
7235	0.38	-1.67E-03

Sum up →

0.311

Perform the carbon footprint of a morning cup of coffee

1. Search for the process number in the ecoinvent tab.

Enter these in column A of the “selected processes” and “carbon footprint” tabs

Process #	Morning coffee robusta Functional unit (FU)= 1 cup of coffee of 0.1 Liter DK electricity
Manufacturing (Materials + Processing)	
	Coffee green beans production robusta (11.5g), RoW
8864	Heat from natural gas for coffee roasting
	Electricity for coffee roasting. DK, Electricity low voltage
2325	Coffee machine (1 unit of 3 kg)
10771	Paper Cup (8g paper)
480	Packaging (capsule 2g alu - aluminium)
Transportation	
18241	Coffee, transport, freight, lorry 16-32 metric ton (3500 km)
18438	Coffee transport, freight, sea, container ship, (4500 km)
Use phase	
17757	Tap water (0.1 L/cup)
0	Electricity hot water (0.12kWh/L) production mix US
0	Stand-by electricity (0.028 kWh/cup)
Disposal	
19319	Aluminium sent to sanitary landfill
19889	Coffee maker as plastic mixture incineration
19681	Paper cup sent to incineration
8864	Avoided heat paper and plastic incineration
0	Avoided electricity paper and plastic incineration
Total	

2. Enter in column F the amount of units (Col.G) of each process per FU, i.e per 1 cup of coffee

Check your units!

Quantity per FU	Unit
FU = 1 cup of coffee	
	#N/A
3.02E-02	MJ
	#N/A
	unit
8.00E-03	kg
2.00E-03	kg
per FU	
0.00E+00	metric ton*km
	metric ton*km
FU = per cup of coffee	
1.00E-01	kg
	#N/A
2.80E-02	#N/A
per FU	
2.00E-03	kg
0.00E+00	kg
	kg
-7.68E-02	MJ
-8.89E-03	#N/A
	-

3. Multiply F x H to get the carbon footprint per FU

Carbon footprint [kgCO2equ/unit]	Carbon footprint [CO2equ./FU]
#N/A	
0.07	2.21E-03
#N/A	#N/A
41.72	0.00E+00
0.72	5.79E-03
9.89	1.98E-02
0.19	0.00E+00
0.01	0.00E+00
0.00	3.09E-05
#N/A	#N/A
#N/A	#N/A
0.02	3.03E-05
2.34	0.00E+00
0.01	0.00E+00
0.07	-5.61E-03
#N/A	#N/A
	#N/A

4. Interpret What are the dominant processes?

Rank from highest to lowest

Sum up